



清華大學

1. 1) $k_p' \left(\frac{W}{L}\right)_p V_{DSATP} \left(-V_{DD} - V_{TP} - \frac{V_{DSATP}}{2}\right) (1 + \lambda_p V_{OL} - \lambda_p V_{DD}) +$

$k_n' \left(\frac{W}{L}\right)_n V_{OL} \left(V_{DD} - V_{TN} - \frac{V_{OL}}{2}\right) (1 + \lambda_n V_{OL}) = 0 \quad (1)$

$\left(\frac{W}{L}\right)_p = \frac{k_n' V_{OL}}{k_p' V_{DSATP}} \cdot \left(\frac{V_{DD} + V_{TP} + \frac{V_{DSATP}}{2}}{V_{DD} - V_{TN} - \frac{V_{OL}}{2}} \right) \cdot \frac{1 + \lambda_n V_{OL}}{1 - \lambda_p (V_{DD} - V_{OL})} = 1.15$

2) 此時 $k_p' \left(\frac{W}{L}\right)_p V_{DSATP} \left(-V_{DD} - V_{TP} - \frac{V_{DSATP}}{2}\right) = -3.6 \times 10^{-6} - 9.6 \times 10^{-5} \text{ A/V}$

$k_n' \left(\frac{W}{L}\right)_n = 4.6 \times 10^{-6} \text{ A/V}^2$, $1 + \lambda_p V_{DD} = 1.25$, $V_{DD} - V_{TN} = 2.07 \text{ V}$

$\lambda_n V_{OL} \ll 1$, $V_{OL} \ll V_{DD} - V_{TN}$, $\lambda_p V_{OL} \ll 1$

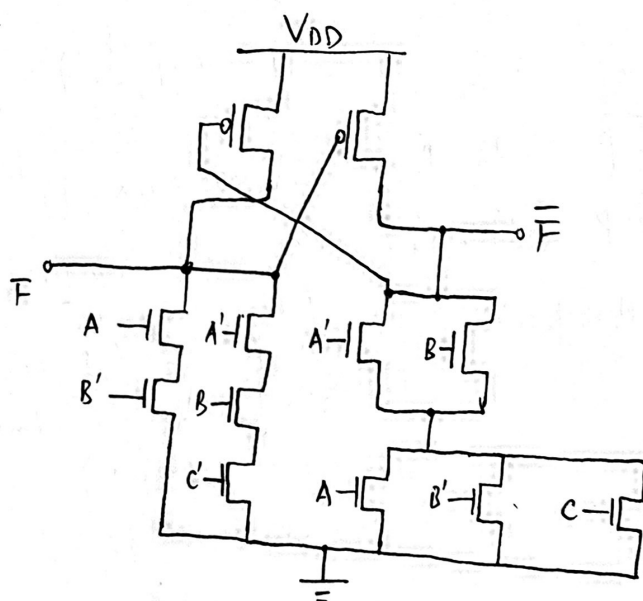
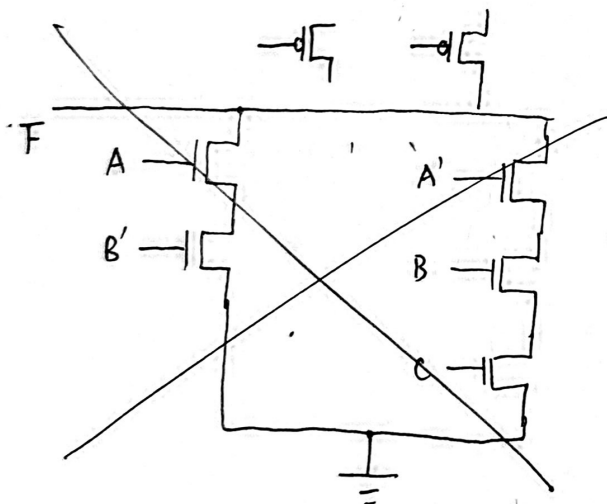
則 (1) 式可變為

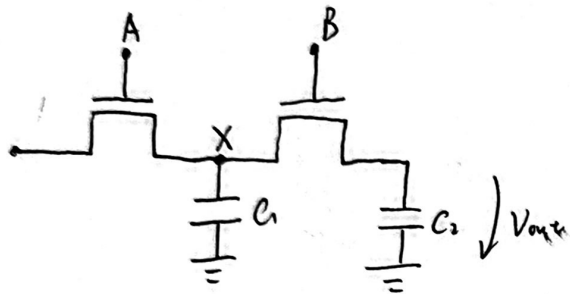
$$V_{OL} = \frac{k_p' \left(\frac{W}{L}\right)_p V_{DSATP} \left(+V_{DD} + V_{TP} + \frac{V_{DSATP}}{2}\right) (1 - \lambda_p V_{DD})}{k_n' \left(\frac{W}{L}\right)_n (V_{DD} - V_{TN})}$$

$$= \frac{+9.6 \times 10^{-5} \times 1.25}{4.6 \times 10^{-6} \times 2.07} = 0.13 \text{ V}$$

$I_{OL} = k_n' \left(\frac{W}{L}\right)_n V_{OL} \left(V_{DD} - V_{TN} - \frac{V_{OL}}{2}\right) (1 + \lambda_n V_{OL}) = 4.6 \times 10^{-6} \times 0.13 \times \left(2.5 - 0.43 - \frac{0.13}{2}\right) (1 + 0.06 \times 0.13)$
 $= 0.12 \text{ mA}$

2. $F = \overline{A} \overline{B} \overline{C} + \overline{A} B \overline{C} + A \overline{B} C + A B C = (A + B + C)(A' + B + C)(A' + B + C') = (A + B + C)(A' + B)$





最终:

$$V_B = V_{DD} = 2.5V$$

$$V_B - V_X = V_{TN}$$

$$C_1 V_X + C_2 V_{out} = C_2 V_{DD}$$

$$\Rightarrow C_1 (V_{DD} - V_{TN}) + C_2 V_{out} = C_2 V_{DD}$$

$$\begin{aligned} \Rightarrow V_{out} &= \left(1 - \frac{C_1}{C_2}\right) V_{DD} + \frac{C_1}{C_2} V_{TN} \\ &= 0.6 \times 2.5V + 0.4 \times 0.5 \\ &= 1.7V \end{aligned}$$

4. a) 只有一个输入是高电平:

$$V_0 = V_{OL,1} \approx \frac{k'_p \left(\frac{W}{L}\right)_p}{k'_n \left(\frac{W}{L}\right)_n} \cdot \frac{V_{DD} + V_{TP} + \frac{V_{DSATP}}{2}}{V_{DD} - V_{TN}} \cdot (1 - \lambda_p V_{DD}) V_{DSATP} = 0.10V$$

四个高电平输入:

$$V_0 = \frac{V_{OL,1}}{4} = 0.025V$$

b) 设 $I_{D,n}$ 是 n 个 NMOS 同时输入高电平时的导通电流

$$\cancel{I_{D,n} = k'_n \left(\frac{W}{L}\right)_n \left(V_{OL,1} - V_{TN}\right)^2} \quad , \quad I_{D,n} = k'_p \left(\frac{W}{L}\right)_p V_{DSATP} \left(V_{DD} + V_{TP} + \frac{V_{DSATP}}{2}\right) (1 - \lambda_p V_{DD})$$

$$= k'_n \left(\frac{W}{L}\right)_n V_{OL,1} \left(V_{DD} - V_{TN} - \frac{V_{OL,1}}{2}\right)$$

$$= 36 \mu A \quad (n \geq 1)$$

$$I_{D,n} = 0A \quad (n=0)$$

$$\overline{P_{stat}} = V_{DD} \overline{I_{stat}} = V_{DD} \cdot \sum_{n=0}^4 C_4^n p^n (1-p)^{4-n} I_{D,n}$$

$$= V_{DD} \sum_{n=0}^4 C_4^n p^n (1-p)^{4-n} I_{D,n}$$

$$= V_{DD} (I_{D,1} - (1-p)^4 I_{D,1}) = V_{DD} I_{D,1} [1 - (1-p)^4] = 2.5V \times 0.36 \mu A \times (1 - 0.625)$$

$$\text{当 } p=0.5 \text{ 时, } \overline{P_{stat}} = 2.5V \times 36 \mu A \times [1 - 0.5^4] = 84.375 \mu W$$

$$\text{当 } p=0.1 \text{ 时, } \overline{P_{stat}} = 2.5V \times 36 \mu A \times [1 - 0.9^4] = 30.951 \mu W$$

c) 仿真结果: 单输入高电平: $V_0 = 105.24mV$

四输入高电平: $V_0 = 25.009mV$

$$\overline{P_{stat}} = 101 \mu W \quad (p=0.5)$$

$$25 \mu W \quad (p=0.1)$$